

MATH5210 SYLLABUS

SPRING 2026

Instructor: Kurt Vinhage

Office: 309 JWB

Office Hours: TBA

Contact. The best way to contact me is through email (vinhage@math.utah.edu). I will respond to emails within 48 hours, but many times sooner.

Course Content. We use a variety of resources and textbook chapters which will be made available as pdf documents. The course will aim to touch on basics of measure theory, functional analysis, and time permitting, ergodic theory. Textbooks used may include:

- Principles of Mathematical Analysis, Rudin
- Topology, Munkres
- Real Analysis, Fitzpatrick & Royden
- Introduction to Dynamical Systems, Brin & Stuck

Homeworks and Quizzes. There will be regular homework assignments. Homeworks will be given roughly weekly (initially Fridays), and announced in-class and on CANVAS.

Homework Policies.

- Homework must be typed in LaTeX.
- Homeworks may be completed in groups of up to four people.
- Homeworks will receive full credit for honest attempts of every problem. Solutions will be posted promptly.

Midterms and Final. Exams are timed and no outside resources will be allowed. There will be one midterm exam on roughly February 27, 2026. The exam will be given in-class and be graded out of 100 points. **The date of the midterm is preliminary and subject to change.**

The final exam will be graded out of 100 points.

Course Grades. The course grades will be calculated based on the following scale:

Homeworks	30% total
Midterm	30%
Final	40%

The letter-grade cutoffs will be determined with the following thresholds. *These are guaranteed thresholds, they may be lowered later.*

Letter	Percent
A	93%
A-	90%
B+	87%
B	83%
B-	80%
C+	77%
C	73%
C-	70%
D	60%
F	50%

Course goals. We aim to cover the following topics over the course of the semester:

- Arzela-Ascoli Theorem
- Banach & Hilbert spaces
- σ -algebras and measures
- Lebesgue measure
- L^p -spaces
- Fourier analysis on $\mathbb{R}$ and $\mathbb{R}/\mathbb{Z}$
- Bernoulli and Markov measures
- Ergodicity and Mixing

AMERICANS WITH DISABILITIES ACT (ADA)

The University of Utah seeks to provide equal access to its programs, services, and activities for people with disabilities.

All written information in this course can be made available in an alternative format with prior notification to the Center for Disability & Access (CDA). CDA will work with you and the instructor to make arrangements for accommodations. Prior notice is appreciated. To read the full accommodations policy for the University of Utah, please see Section Q of the Instruction & Evaluation regulations.

In compliance with ADA requirements, some students may need to record course content. Any recordings of course content are for personal use only, should not be shared, and should never be made publicly available. In addition, recordings must be destroyed at the conclusion of the course.

If you will need accommodations in this class, or for more information about what support they provide, contact:

Center for Disability & Access 801-581-5020 disability.utah.edu Third Floor, Room 350 Student Services Building 201 S 1460 E Salt Lake City, UT 84112
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Safety at the U. The University of Utah values the safety of all campus community members. You will receive important emergency alerts and safety messages regarding campus safety via text message. For more safety information and to view available training resources, including helpful videos, visit safeu.utah.edu.

To report suspicious activity or to request a courtesy escort, contact:

Campus Police & Department of Public Safety
 801-585-COPS (801-585-2677)
dps.utah.edu
 1735 E. S. Campus Dr.
 Salt Lake City, UT 84112

ADDRESSING SEXUAL MISCONDUCT

Title IX makes it clear that violence and harassment based on sex and gender (which includes sexual orientation and gender identity/expression) is a civil rights offense subject to the same kinds of accountability and the same kinds of support applied to offenses against other protected categories such as race, national origin, color, religion, age, status as a person with a disability, veteran's status, or genetic information.

If you or someone you know has been harassed or assaulted, you are encouraged to report it to university officials:

Office of Equal Opportunity and Title IX
 801-581-8365
oeo.utah.edu
 135 Park Building
 201 Presidents' Cir.
 Salt Lake City, UT 84112

Office of the Dean of Students
 801-581-7066
deanofstudents.utah.edu
 270 Union Building
 200 S. Central Campus Dr.
 Salt Lake City, UT 84112

To file a police report, contact:

Campus Police & Department of Public Safety
 801-585-COPS (801-585-2677)
dps.utah.edu
 1735 E. S. Campus Dr.
 Salt Lake City, UT 84112

If you do not feel comfortable reporting to authorities, the U's Victim-Survivor Advocates provide free, confidential, and trauma-informed support services to students, faculty, and staff who have experienced interpersonal violence.

To privately explore options and resources available to you with an advocate, contact:

Center for Campus Wellness
 801-581-7776
wellness.utah.edu
 350 Student Services Building
 201 S. 1460 E.
 Salt Lake City, UT 84112

ACADEMIC MISCONDUCT

It is expected that students comply with University of Utah policies regarding academic honesty, including but not limited to refraining from cheating, plagiarizing, misrepresenting one's work, and/or inappropriately collaborating. This includes the use of generative artificial intelligence (AI) tools without citation, documentation, or authorization. Students are expected to adhere to the prescribed professional and ethical standards of the profession/discipline for which they are preparing. Any student who engages in academic dishonesty or who violates the professional and ethical

standards for their profession/discipline may be subject to academic sanctions as per the University of Utah's Student Code: Policy 6-410: Student Academic Performance, Academic Conduct, and Professional and Ethical Conduct.

Plagiarism and cheating are serious offenses and may be punished by failure on an individual assignment, and/or failure in the course. Academic misconduct, according to the University of Utah Student Code:

“...Includes, but is not limited to, cheating, misrepresenting one's work, inappropriately collaborating, plagiarism, and fabrication or falsification of information...It also includes facilitating academic misconduct by intentionally helping or attempting to help another to commit an act of academic misconduct.”

For details on plagiarism and other important course conduct issues, see the U's Code of Student Rights and Responsibilities.